

Interstitial IgG antibody pharmacokinetics assessed by combined *in vivo*- and physiologically-based pharmacokinetic modelling approaches

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Key points

- For therapeutic antibodies, total tissue concentrations are frequently reported as a lump sum measure of the antibody in residual plasma, interstitial fluid and cells. In terms of correlating antibody exposure to a therapeutic effect, however, interstitial pharmacokinetics might be more relevant.
- In the present study, we collected total tissue and interstitial antibody biodistribution data in mice and assessed the composition of tissue samples aiming to correct total tissue measurements for plasma and cellular content.
- All data and parameters were integrated into a refined physiologically-based pharmacokinetic model for monoclonal antibodies to enable the tissue-specific description of antibody pharmacokinetics in the interstitial space.
- We found that antibody interstitial concentrations are highly tissue-specific and dependent on the underlying capillary structure but, in several tissues, they reach relatively high interstitial concentrations, contradicting the still-prevailing view that both the distribution to tissues and the interstitial concentrations for antibodies are generally low.

Abstract For most therapeutic antibodies, the interstitium is the target space. Although experimental methods for measuring antibody pharmacokinetics (PK) in this space are not well established, thus making quantitative assessment difficult, the interstitial antibody concentration is assumed to be low. In the present study, we combined direct quantification of antibodies in the interstitial fluid with a physiologically-based PK (PBPK) modelling approach, with the aim of better describing the PK of monoclonal antibodies in the interstitial space of different tissues. We isolated interstitial fluid by tissue centrifugation and conducted an antibody biodistribution study in mice, measuring total tissue and interstitial concentrations in selected tissues. Residual plasma, interstitial volumes and lymph flows, which are important PBPK model parameters, were assessed *in vivo*. We could thereby refine the PBPK modelling of monoclonal antibodies, better interpret antibody biodistribution data and more accurately predict their PK in the different tissue spaces. Our results indicate that, in tissues with discontinuous capillaries (liver and spleen), interstitial concentrations are reflected by the plasma concentration. In tissues with continuous capillaries (e.g. skin and muscle), ~50–60% of the plasma concentration is found in the interstitial space. In the brain and kidney, on the other hand, antibodies are restricted to the vascular space. Our data may significantly impact the interpretation of biodistribution data of monoclonal antibodies and might be important when relating measured concentrations to a therapeutic effect. By contrast to the view that the antibody distribution to the interstitial space is limited, using direct measurements and model-based data interpretation, we show that high antibody interstitial concentrations are reached in most tissues.

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Abbreviations $C_{\max}$, maximal concentration; f_{Vecis} , tissue extracellular volume fraction; f_{Vintis} , tissue interstitial volume fraction; f_{Vresplis} , tissue residual plasma volume fraction; FcRn, neonatal Fc receptor; HSA, human serum albumin; IL-17, interleukin 17; PBPK, physiologically based pharmacokinetic; PK, pharmacokinetics; PKPD, pharmacokinetic/pharmacodynamic.

Introduction

The importance of therapeutic antibodies in drug therapy has increased steadily over recent years. The pharmacokinetics (PK) of large molecules differs in many aspects from small molecule drugs, as discussed in several recent reviews (Jones *et al.* 2013; Ferl *et al.* 2016; Wan, 2016). Nonetheless, the distribution of antibodies in the subcompartments (vascular, interstitial and cellular space) of different tissues is still not well-established in quantitative terms because experimental measurements at the subcompartmental level of tissues are challenging. Therefore, most PK data for large molecules are reported as total tissue concentrations; however, this may not represent the effects driving concentration and, instead, comprise a mixture of vascular, intracellular and interstitial concentrations (Danhof *et al.* 2007; Mouton *et al.* 2008; Mariappan *et al.* 2013). Notable differences in concentration levels are expected between the tissue subcompartments (Lobo *et al.* 2004; Wang *et al.* 2008). Many therapeutic antibodies bind to targets on the cell surface and thus induce their therapeutic effect within the interstitial space of a tissue (i.e. the biophase) (Boswell *et al.* 2012). Therefore accurate measurements or predictions of tissue interstitial concentrations are critical for estimating how much of the drug reaches the therapeutic target and evaluating the pharmacokinetic/pharmacodynamic (PKPD) properties of therapeutic antibodies during drug development (Danhof *et al.* 2007; Mariappan *et al.* 2013).

So far, there is no broadly accepted gold standard method for experimentally assessing the antibody concentration in the interstitial space (i.e. at the target site) (Wiig & Swartz, 2012). If total tissue concentrations are measured by an enzyme-linked immunoabsorbent assay or radiolabelling, subsequent corrections for residual plasma contamination and interstitial volume fractions of the underlying tissue might be applied to estimate the extravascular and interstitial tissue concentrations, respectively (Garg, 2007; Fronton *et al.* 2014). Alternatively, physiologically based pharmacokinetic (PBPK) models provide a robust tool for predicting the PK in the different tissue subcompartments and can be used further to evaluate PK properties of compounds in development and to scale between different species or patient populations. They mathematically describe the distribution of compounds throughout the body based on physiological processes, anatomical structures and

physicochemical drug properties. Yet, because of the lack of physiological and quantitative knowledge surrounding driving processes for the biodistribution of large molecules, these models still contain a number of unknown physiological parameters that are required as inputs. This is also reflected by the diverse structure and parameterization of already published large molecule PBPK models (Covell *et al.* 1986; Baxter *et al.* 1994; Ferl *et al.* 2005; Garg & Balthasar, 2007; Davda *et al.* 2008; Urva *et al.* 2010; Chen & Balthasar, 2012; Shah & Betts, 2012; Jones *et al.* 2013; Fronton *et al.* 2014). The further development and validation of large molecule PBPK models therefore requires additional research to inform our understanding of the underlying distribution processes.

In the present study, our objective was to investigate the PK of monoclonal antibodies in the interstitial space of individual tissues. Such knowledge would allow us to clarify how much of the administered therapeutic dose will reach the target site in a specific tissue. We report new biodistribution data, including measured interstitial concentrations in selected tissues, and parameter values forming the basis of a PBPK modelling approach, thereby providing new knowledge on the tissue distribution and effect-driving concentrations of monoclonal antibodies. Our new findings are integrated into a PBPK model framework, enabling us to make antibody PK predictions at the target site in different tissues. Our data suggest a high antibody exposure in the interstitial space of most organs, except the brain and kidney and we show that the often reported total tissue concentrations are much lower for many tissues compared to their interstitial concentrations. We demonstrate that an accurate assessment and interpretation of correction factors and input parameters is pivotal for the assessment of antibody PK. Importantly, our results clearly indicate that high interstitial concentrations can be achieved for antibodies in most tissues, which contradicts the prevalent assumption that the therapeutic antibody concentrations in this space are low.

Methods

Ethical approval

All *in vivo* studies were carried out in accordance with the regulations of the Norwegian State Commission for

Laboratory Animals in agreement with the European Convention for the Protection of Vertebrate Animals used for Experimental and Other Scientific Purposes and Council of Europe (ETS 123) and were approved by the AAALAC International Accredited Animal Care and Use Program at University of Bergen. The study is also confirmed to be compliant with the ethical principles under which *The Journal of Physiology* operates and adheres with the Animal Ethics Checklist presented in the editorial by Grundy (Grundy, 2015).

Animal studies

FVB/NhanHsd mice supplied by Envigo (An Venray, The Netherlands) were used for all *in vivo* measurements. For every study, both sexes were used (total of 34 females and 26 males). The body weights ranged from 19 to 34 g, with a mean of 23 g, and the ages of the animals ranging from 8 to 12 weeks. The mice were fed *ad libitum*. During invasive experimental procedures, the animals were anaesthetized with 1.5% isoflurane (IsoFlo® Vet 100%; Abbott Laboratories Ltd, Maidenhead, UK) in 100% O₂. The depth of anaesthesia was monitored by testing the withdrawal reflexes (paw). During anaesthesia, the body temperature was kept stable using a servo-controlled heating pad and rectal probe. At the end of the experiment, animals were killed by cervical dislocation.

Radiolabelled probes

We used various radiolabelled compounds in our *in vivo* studies. ⁵¹Cr-EDTA (produced by GE Healthcare Limited; delivered by IFE, Institute for Energy Technology, Kjeller, Norway) was utilized for assessment of extracellular spaces. Human serum albumin (HSA) and anti-interleukin 17 (IL-17) IgG were labelled with ¹²⁵I using iodogen as described in detail previously (Wiig *et al.* 2005). In summary, a solution of 5 mg of 1,3,4,6-tetrachloro-3 α ,6 α -diphenylglycouril (product number T0656; Sigma-Aldrich Co., St Louis, MO, USA) in 5 ml of chloroform was prepared and 0.1 ml of this was transferred into a 1.8-ml Nunc vial (Nunc-Kamstrup, Roskilde, Denmark). Chloroform was evaporated under nitrogen forming water-insoluble Iodo-Gen in the Nunc vial. Next, 1.5 mg of compound was added to 1 ml of 0.05 M PBS with 15 μ l of 0.01 M NaI and 10 MBq ¹²⁵I (Institute for Energy Technology). The vial was gently mixed for 10 min before the solution was removed. The tracer solution was dialysed against 1 litres of 0.9% of saline and 0.02% azide to remove unincorporated isotope. The labelled probes were additionally purified prior to each i.v. injection using a 40 kDa cut-off spin filter (Amicon Corp., Danvers, MA, USA). Separate terminal plasma and urine samples were collected during the study and tested for free iodine and purity using spin filters and HPLC.

Biodistribution study

Concentrations of ¹²⁵anti-IL-17 IgG were measured in plasma and 11 tissues (adipose, bone, brain, gut, heart, kidney, liver, lung, muscle, skin and spleen) at 10 time points with three mice at each time point. Sampling times were 10, 30 and 90 min; 3, 6 and 12 h; and 1, 2, 5 and 15 days, with the focus being on early time points to resolve the early tissue distribution phase. A dose of 10 mg kg⁻¹ was administered i.v. through the tail vein. For terminal tissue sampling, mice were anaesthetized; a blood sample was retrieved from the tail vein using heparinized glass capillaries, followed by death. Plasma was isolated by centrifugation of the blood sample for 10 min at 1000 g. Harvested tissue samples were blotted dry on a paper tissue to remove surface blood. Separate samples for skin and muscle were obtained to isolate interstitial fluid by tissue centrifugation. Samples were weighed, transferred to vials and radioactivity was determined in a gamma-counting system (Wallac Wizard 1470 gamma counter; PerkinElmer, Waltham, MA, USA). An additional blood sample was taken at several time points and filtered through a spin filter (40-kDa cut-off) to determine the amount of free iodine in the system. The amount of drug was assessed based on specific activity of the probe and correction for radioactive decay. Drug content in tissues is reported per gram wet weight of the respective tissue. The tissue sample volumes (V_{tis}) were calculated based on the measured sample weights:

$$V_{tis} = \frac{\text{Weight}}{\text{Density}} \quad (1)$$

A density of 1 g cm⁻³ was assumed for all tissues, except 0.92 and 1.3 g cm⁻³ for adipose and bone, respectively (Brown *et al.* 1997; Valentin, 2002). This allowed the total tissue concentrations to be derived.

Tissue centrifugation

Interstitial concentrations. A preliminary study was performed to test the isolation of native interstitial fluid using the tissue centrifugation technique (Wiig *et al.* 2003) from 11 tissues (adipose, bone, brain, gut, heart, kidney, liver, lung, muscle, skin and spleen) and evaluate the plasma admixture and dilution by intracellular fluid in the centrifugate of each tissue. Based on the data obtained, muscle and skin were selected to establish interstitial PK during the biodistribution study because these tissues allowed the collection of relatively pure interstitial fluid with limited plasma and intracellular fluid contamination. To avoid evaporation, all procedures were performed in a humidity chamber (98% relative humidity). Tissue samples were placed on a mesh in an Eppendorf tube and centrifuged at low speed of 424 g for 10 min. After centrifugation, the tubes were immediately

transferred back to the humidity chamber where isolated fluid samples at the bottom of the tube were transferred into a tube containing 500 μ l of saline and counted in the gamma counter to measure the 125 anti-IL-17 IgG content in the centrifugate.

Tissue volume fractions

Extracellular space by $^{51}\text{Cr-EDTA}$. The time needed for tracer equilibration was tested in preliminary studies. Under anaesthesia, two 1-cm incisions penetrating the skin and muscle layer were made lateral from the spine and below the rib cage, and the kidney pedicles were tied off to prevent tracer excretion. Following wound closure, $^{51}\text{Cr-EDTA}$ (~ 16.6 kBq in 100 μ l) was injected i.v. into the tail vein of seven mice. Serial blood sampling from the tail vein after 30, 60 and 90 min revealed that tracer levels in plasma were not different at these time points. Although this suggests that the tracer was equilibrated already at 30 min, we chose the 60-min equilibration time for the extracellular tracer to ascertain complete equilibration. In a group of six mice, $^{51}\text{Cr-EDTA}$ was injected as described before. The mice were kept under anaesthesia for 60 min before a terminal blood sample was withdrawn and the animals were killed. Subsequently, adipose tissue, bone, brain, gut, heart, kidney, liver, lung, muscle, skin and spleen were harvested. Surface blood was removed by briefly blotting tissues on a paper tissue. All samples were weighed, transferred to counting tubes and counted in the gamma counter. The tissue extracellular volume fraction ($f_{\text{Vec}_{\text{tis}}}$) was calculated as:

$$f_{\text{Vec}_{\text{tis}}} = \frac{{}^{51}\text{Cr counts/g tissue}}{{}^{51}\text{Cr counts/ml plasma}} \quad (2)$$

Residual plasma space by $^{125}\text{I-HSA}$. After anaesthesia, six mice received an i.v. bolus injection of $^{125}\text{I-HSA}$ (~ 17 kBq in 100 μ l) with a 5-min distribution time, which was sufficient to allow distribution in but not extravasation from plasma (with few exceptions, as described below) before blood sampling and death. Thereafter, tissues were harvested, blotted dry and transferred to the gamma counter. Importantly, no other measures for removing additional blood were employed. Residual plasma fractions ($f_{\text{Vrespla}_{\text{tis}}}$) in harvested tissues were calculated as:

$$f_{\text{Vrespla}_{\text{tis}}} = \frac{{}^{125}\text{I counts/g tissue}}{{}^{125}\text{I counts/ml plasma}} \quad (3)$$

Interstitial volume fractions. We used our measured volume fractions to derive a tissue interstitial volume fraction ($f_{\text{Vint}_{\text{tis}}}$) based on the relationship:

$$f_{\text{Vint}_{\text{tis}}} = f_{\text{Vec}_{\text{tis}}} - f_{\text{Vrespla}_{\text{tis}}} \quad (4)$$

Lymph flow measurements

To measure the lymph flow in muscle and skin, Alexa 680-labelled macromolecular tracers (Invitrogen, Carlsbad, CA, USA) were injected intradermally and intramuscularly in the hind paw and thigh of mice respectively. Washout rates of the tracer from the injection site were assessed by optical imaging using the Optix MX system (West Medica Produktions- und Handels- GmbH, Vienna, Austria) to measure the fluorescence signal. The rates were computed based on the monoexponential reduction of the fluorescence signal as described in detail previously (Karlsen *et al.* 2012). In brief, we fluorescently labelled BSA, anti-IL-17 IgG antibody and a Triple-A mutant [neonatal Fc receptor (FcRn) non-binding] antibody with the near infrared Alexa 680 fluorophore and an antibody labelling kit (SAVI Rapid Antibody Labelling Kit; Invitrogen). The three different macromolecular tracers were employed to check for a possible effect of molecular weight (BSA *vs.* anti-IL-17 IgG) and FcRn binding (anti-IL-17 IgG *vs.* FcRn non-binding IgG) upon removal from the interstitial space. Volumes of 0.5 μ l of tracer were injected with a 34-gauge Hamilton syringe. After a 60-min distribution phase, five measurements were taken at intervals of 1 h. Animals were anaesthetized during imaging but were awake and freely moving inbetween the measurements to assess washout of normally active animals, knowing that immobility reduces lymph flow (Lindena *et al.* 1986; Modi *et al.* 2007). For each tracer and tissue, the washout was assessed in six individuals. Optix Optiview software was used to analyse the images. Flow values were subsequently derived by multiplying removal rates from the interstitial volume of the respective tissue:

$$L_{\text{tis}} = f_{\text{Vint}_{\text{tis}}} \times V_{\text{tis}} \times k \quad (5)$$

where L_{tis} is the tissue lymph flow, f_{Vint} is the interstitial volume fraction, V_{tis} is the tissue volume and k is the measured tracer removal rate.

Corrections of total tissue concentrations

During the biodistribution experiments, plasma and total tissue concentrations were measured. Extravascular tissue concentrations were derived by subtracting the amount of antibody in the residual plasma from the measured total amount found in tissue and dividing what remained by the extravascular tissue volume (Garg, 2007; Fronton *et al.* 2014):

$$C_{\text{ev}} = \frac{C_{\text{tis}} \times V_{\text{tis}} - C_{\text{pla}} \times V_{\text{respla}_{\text{tis}}}}{V_{\text{tis}} - V_{\text{respla}_{\text{tis}}}} \quad (6)$$

where C_{ev} is the tissue extravascular concentration, C_{tis} is the experimentally assessed total tissue concentration, C_{pla} is the plasma concentration and V_{tis} is the experimentally measured total tissue sample volume. $V_{\text{respla}_{\text{tis}}}$

is the residual plasma volume in the respective tissue sample and is determined based on the corresponding measured residual plasma fraction multiplied by the total sample volume. Residual plasma fractions utilized for correction were based on the ^{125}I -HSA distribution. Under the same assumptions, we additionally used the measured amount of anti-IL-17 IgG in tissues after 10 min in the biodistribution study for residual plasma corrections.

Statistical analysis

Measured data are reported as the mean $\pm$ SD. For comparison of groups, a two-tailed Student's *t* test or one-way ANOVA was conducted. $P \leq 0.05$ was considered statistically significant. Analyses were performed in PRISM, version 6.0 (GraphPad Software Inc., San Diego, CA, USA).

PBPK modelling

We integrated and evaluated the newly measured biodistribution data and parameter values in a PBPK modelling approach. The model was coded in the Simbiology toolbox in MATLAB R2016a (MathWorks Inc., Natick, MA, USA) and was used to fit total tissue concentrations and, where available, interstitial concentrations. Tissue distribution, lymph flow (except for the muscle and skin), systemic plasma volume, systemic clearance and interstitial accessible volume for the skin and muscle were estimated using the *fminsearch*-bound and *lsqnonlin* algorithms, whereas other parameters were fixed to reported values (Shah & Betts, 2012). The measured physiological parameter values (i.e. residual plasma- and interstitial volume fractions and lymph flow in the skin and muscle) were utilized as input parameters for the model and limited to a range of measured values $\pm 20\%$ during the estimation to permit variability. The presented model integrated only well-established or required parameters and mechanisms to describe our data and allow prediction of the concentrations in the different tissue subcompartments. For this analysis, wherever possible, we avoided including parts into the model structure where no experimental data were available and which were not directly affecting our research question (e.g. detailed endosomal compartment, FcRn receptor, target binding, etc.). Furthermore, because all major organs were sampled and included within the model structure, making up for 92% of the body weight and 95% of the cardiac output. Therefore, no additional carcass or rest of body compartment was included. This is not expected to cause a significant impact or bias in terms of amount of antibody distributed to the included organs.

Tissue distribution space, inflow and removal of antibodies were modelled differently for the individual tissues based on their capillary structure. Tissues with continuous

capillaries provide a distinct separation of vascular and interstitial space for macromolecules. Therefore, tissue distribution (representing a lump sum of convective flow and reflection coefficient) and removal (i.e. lymph flow) were estimated. Tissues with especially tight and size-selective capillaries (blood–brain barrier and glomerular filter of the kidneys) were modelled solely by the vascular space because macromolecules are assumed to be restricted here and the measured amount in tissue is entirely explained by the extent of the residual plasma content. Thus, no distribution to the interstitial space or lymph removal was accounted for with these tissues. In the case of tissues with discontinuous capillaries, the vessel wall does not provide a clear separation of vascular and interstitial space for IgGs and the two spaces equilibrate rapidly; therefore, the interstitial concentration is assumed to reach plasma concentration levels within a few minutes. As such, in these tissues, the distribution space was modelled as one lumped space containing the vascular and interstitial compartment. Antibody drugs enter this space by arterial blood flow and are removed by venous blood and lymph flow that, however, cannot be distinguished based on the data. An exclusion volume in the interstitial space of the skin and muscle was estimated based on the measured interstitial volumes and PK data. Tissue-intrinsic clearances were integrated based on previously published contribution of tissue clearances to the total plasma clearance (Eigenmann *et al.* 2017). A schematic representation of the model is provided in Fig. 1 and the model equations are presented in the Appendix. After fitting the PBPK model to the biodistribution data and parameter estimation, the model was utilized to simulate the PK in the tissue subcompartments of the individual tissues. The PBPK model source code (Simbiology file) and a MATLAB script (enabling the running and plotting of the simulations based on the model and parameter values presented in the present study) are provided in the Supporting information (Data S1–4).

Results

Antibody biodistribution study

To assess the normal antibody distribution, we first conducted an anti-IL-17 IgG biodistribution study in normal FvB mice. The biodistribution data, including plasma and tissue PK profiles, are shown in Fig. 2. The measured concentrations represent the total tissue concentrations. After an i.v. dose of 10 mg kg^{-1} , a plasma PK biphasic profile was observed. The maximal concentration (C_{max}) was $215 \mu\text{g ml}^{-1}$ followed by a fast decline in concentration, which indicated the tissue distribution phase. Total tissue concentrations were mostly more than one order of magnitude lower than the plasma

concentration. Well-perfused organs such as the heart, kidney, liver, lung and spleen had generally higher total concentrations, whereas lower concentrations were determined in adipose, bone and muscle. Over the time course of the entire study, negligible free iodine was measured ($<0.4\%$) within the system.

Tissue centrifugation: interstitial concentrations

To measure interstitial concentrations in the skin and muscle and compare them with the respective total tissue PK, we isolated native interstitial fluid from the muscle and skin by tissue centrifugation and directly assessed anti-IL-17 IgG interstitial concentrations. In the centrifugate, the plasma content and intracellular dilution was assessed by ^{125}I -albumin and ^{51}Cr -EDTA centrifugate-to-plasma ratios. Residual plasma fractions of 0.065 and 0.056 were measured in the centrifugate for skin and muscle, respectively. ^{51}Cr -EDTA tissue-to-plasma

ratios of ~ 1.12 and ~ 0.79 were found for the skin and muscle, respectively, indicating slight contamination by intracellular fluid in the muscle centrifugate. The biodistribution data in the centrifugate were corrected for these factors. The interstitial PK profiles for both tissues are depicted in Fig. 3. In each, a C_{max} of $\sim 45 \mu\text{g ml}^{-1}$ was observed, corresponding to $\sim 50\%$ of the plasma concentration after reaching C_{max} . This indicates high interstitial exposure for antibodies in the skin and muscle. Compared to the measured, corresponding total tissue concentrations, the interstitial concentration is >10 times or >3 times higher for muscle and skin, respectively.

Tissue volume fractions

In a next step, we assessed residual plasma and interstitial fluid per gram of tissue to attribute the total amount of measured drug in the tissue sample to the various tissue subcompartments. Residual plasma and extracellular

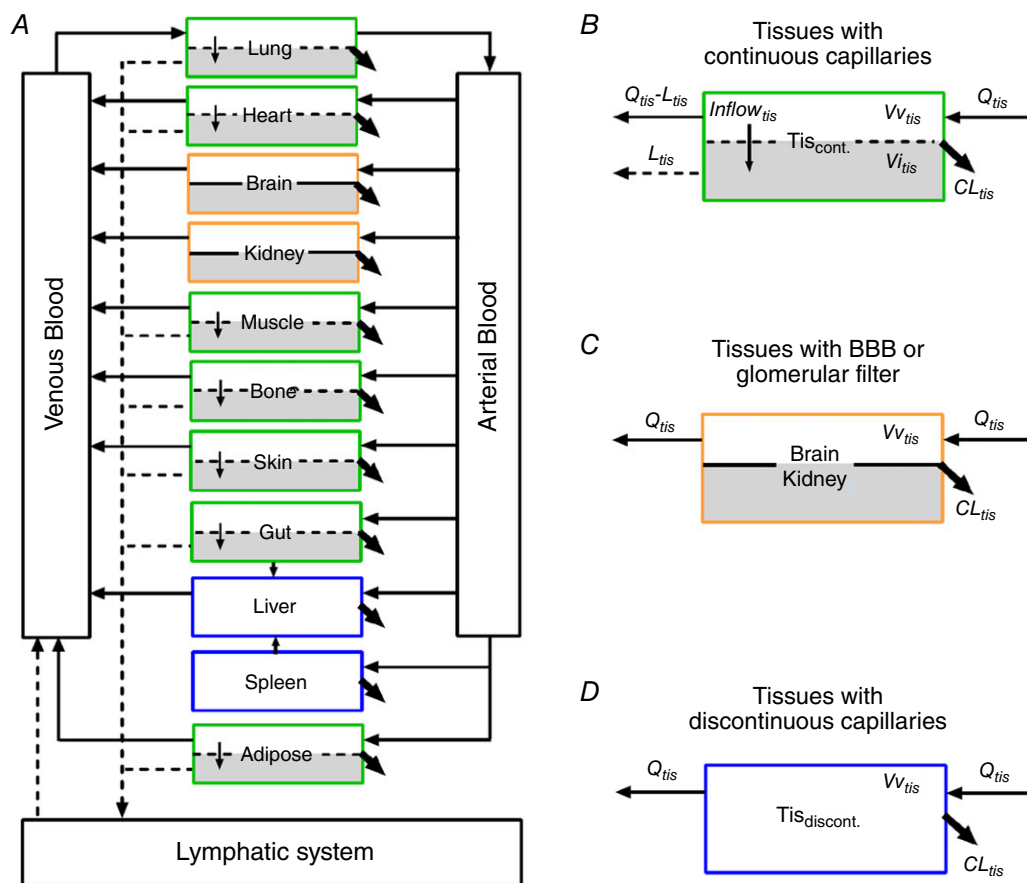


Figure 1. Schematic PBPK model structure

A, schematic model structure. Solid arrows depict blood flow and dashed arrows indicate exit from the tissue interstitial space (grey) by the lymph flow (where estimable). The tissues are modelled depending on their vasculature and are depicted again separately in the subplots, including the respective tissue-specific model parameters. B, tissues with continuous capillaries (green). C, tissues with capillaries largely impenetrable for IgGs [i.e. brain and kidney (orange)]. D, tissues with discontinuous capillaries (blue). The tissue-specific model parameters are: arterial plasma flow (Q_{tis}), lymph flow (L_{tis}), intrinsic clearances (CL_{tis}), residual plasma volume (VV_{tis}) and interstitial volume (Vi_{tis}). Venous blood flow is defined by ($Q_{tis} - L_{tis}$).

volumes were established by tracer distribution, whereas interstitial volumes could be derived by subtracting the residual plasma from the total extracellular space. These volume fractions are important input parameters in the PBPK model and are necessary to more accurately describe distribution to and within the individual tissues.

Extracellular tissue volumes were assessed using a ^{51}Cr -EDTA tracer and by measuring its distribution volumes in tissues 60 min after injection. Extracellular

volumes were derived for all tissues and individual volumes, and the mean and SD are provided in Fig. 4A. The highest mean extracellular volume fraction of 0.45 was found for the skin. No measurements are available for brain (blood–brain barrier) and kidneys (tied off because of excretion). Residual plasma fractions assessed from the 5-min distribution space of ^{125}I -HSA are summarized in Fig. 4B. High residual plasma volumes were found for the lung, kidney, liver, heart and spleen, whereas they were lower in the bone, gut, skin brain, muscle and adipose.

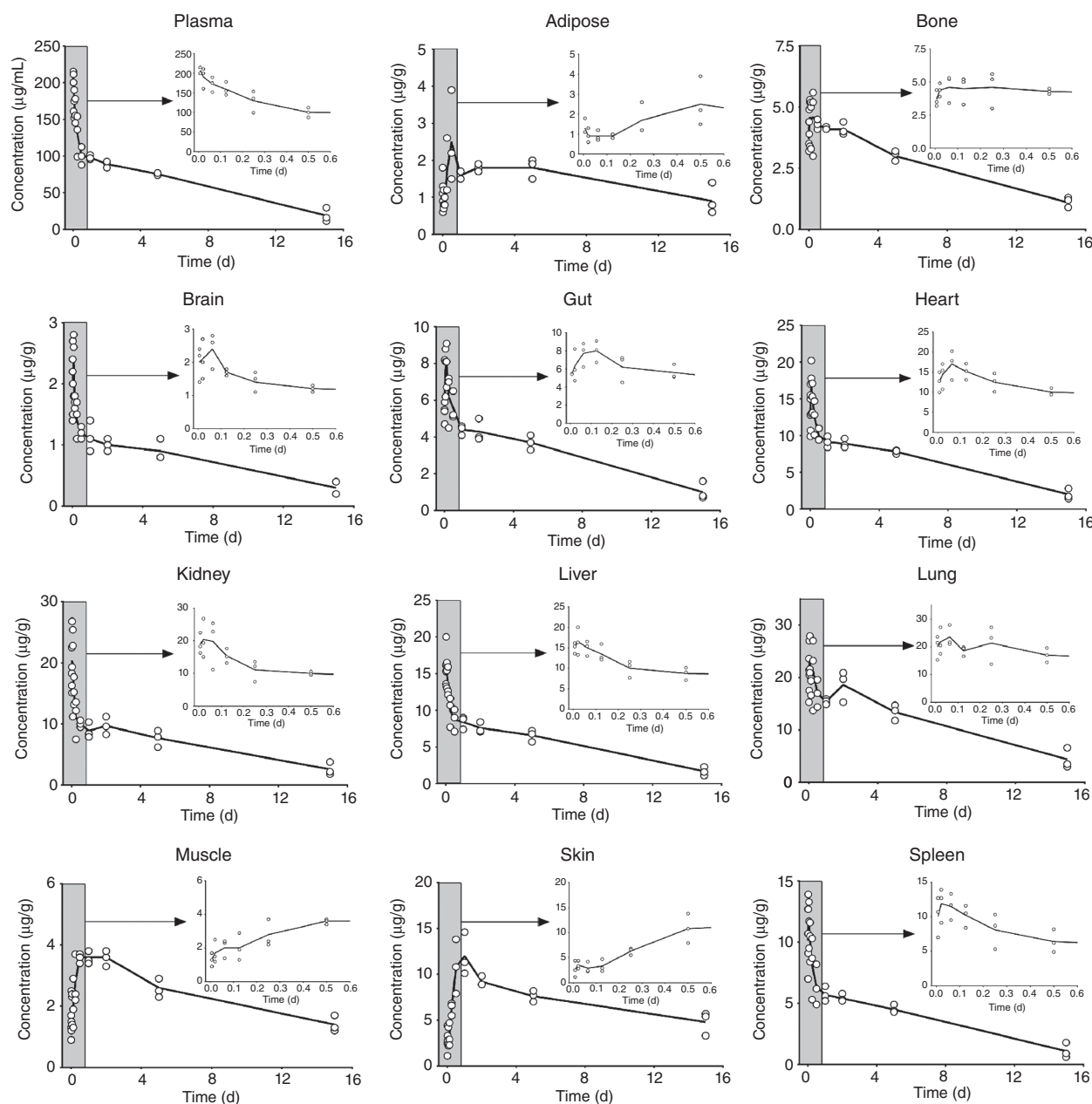


Figure 2. Individual (open circles) and mean (continuous lines) anti-IL-17 IgG total concentrations in plasma and the 11 tissues harvested at 10 time points over 15 days

Each profile is a composite profile (i.e. each individual data point was measured in a different mouse). Also shown in each subplot is a detailed magnified plot of the initial phase (0.6 days) of the concentration–time experiment.

Of note, however, in tissues with leaky, discontinuous capillaries (i.e. liver, spleen and bone marrow), tracer probably extravasated, therefore resulting in an overestimation of the local plasma volume.

We calculated the interstitial space volume in tissues by subtracting the residual plasma from the extracellular fluid fraction. As is evident from Table 1, the skin interstitial volume fraction of 0.431 was by far the highest. By contrast, the lowest interstitial volume fraction of 0.093 was found in adipose tissue. In other tissues, corresponding volume fractions were in the range 0.12–0.24. Overall, the fractional volumes (extracellular, residual plasma and interstitial) could be measured with strong precision. Exceptionally, adipose tissue had a rather high coefficient of variation (42.9–47.8%). All volume fractions and coefficients of variation are summarized in Table 1.

Corrections of total tissue concentrations

Extravascular tissue concentrations derived by subtracting the amount of antibody in the residual plasma from the total tissue concentrations are reported in Figs 5 and 6. The impact is portrayed in terms of residual plasma contamination in the different tissues on the measured antibody content in total tissue samples, as well as the importance of accounting for it when interpreting total tissue measurements. The blue shaded areas are derivative of correcting for residual plasma fractions as measured based on tissue distribution of labelled HSA and anti-IL-17 IgG after 5 and 10 min of distribution, respectively. As is evident from Fig. 5, there was a more profound influence of residual plasma volume correction in highly perfused tissues, such as the lung and heart, vs. the lesser perfused adipose, gut, muscle and skin.

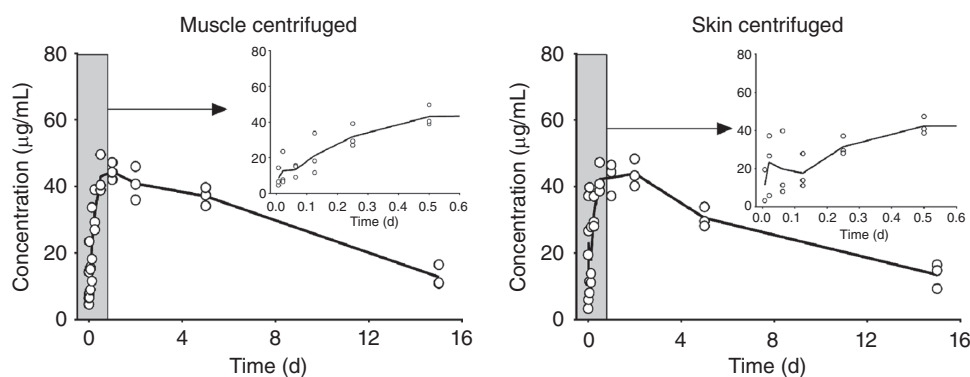


Figure 3. Concentration of anti-IL-17 IgG in the isolated fluid of the muscle and skin as a function of time after injection

Open circles indicate individual measurements (three mice per time point) and the solid curve is the mean profile. A detailed magnified plot of the initial phase (0.6 days) of the concentration-time experiment is shown in both subplots.

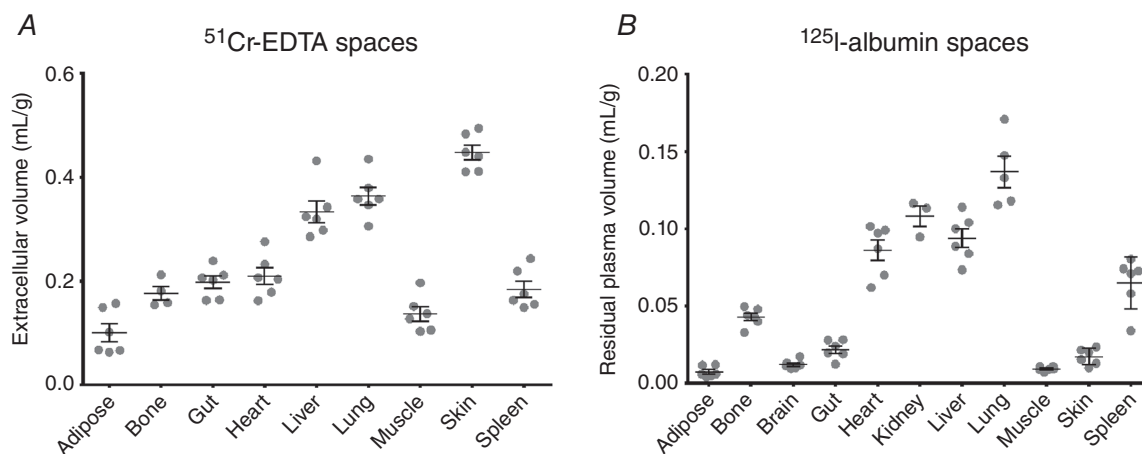


Figure 4. Tissue extracellular- and residual plasma fraction measured in radiotracer studies

A, measured ^{51}Cr -EDTA spaces for different tissues in six mice. Individual volumes are reported per gram wet weight and the mean $\pm$ SD is shown for each tissue. B, depiction of residual plasma fractions remaining in the tissues after harvesting. Individual measurement data, with the mean $\pm$ SD given for each tissue.

Table 1. Subcompartmental volume fractions in tissues

Tissue	Adipose	Bone	Brain	Gut	Heart	Kidney	Liver	Lung	Muscle	Skin	Spleen
f_{ECV}	0.101	0.176	–	0.197	0.210	–	0.333	0.364	0.137	0.448	0.184
CV (%)	42.9	15.0	–	14.9	19.3	–	15.6	11.7	25.3	7.9	20.8
f_{ResPla}	0.007	0.043	0.012	0.022	0.086	0.108	0.094	0.137	0.009	0.017	0.065
CV (%)	47.8	14.1	22.7	27.8	19.2	10.9	15.7	16.7	15.5	30.6	26.0
f_{Vint}	0.093	0.133	–	0.175	0.123	–	0.239	0.227	0.127	0.431	0.119
CV (%)	46.2	20.3	–	17.1	35.8	–	22.6	21.1	27.6	8.4	35.3

Showing measured extracellular (f_{ECV}), residual plasma volume fractions (f_{ResPla}) and derived interstitial volume fractions (f_{Vint}) in ml g^{-1} tissue presented as mean and coefficient of variation (CV) for all 11 tissues.

The same correction approach was applied for bone, brain, kidney, liver and spleen (Fig. 6), although, in these tissues, the entire amount of the drug appears to be in the measured ^{125}I -albumin- or very early anti-IL-17 IgG distribution spaces. After subtraction of the amount of drug within these spaces, the anticipated concentration range approximates towards zero.

Lymph flow in muscle and skin

Lymph flow is a critical parameter for antibody biodistribution because it represents the exit route for macromolecules from the interstitial space back to the plasma. This parameter was assessed in muscle and skin based on near-infrared labelled BSA, anti-IL-17 IgG and FcRn non-binding IgG. We investigated whether there was an influence of molecular size (between 66.4 and 150 kDa) and of FcRn binding on the macromolecular washout from the tissue interstitial space. A size-dependent hindrance in accessing lymph vessels would appear as a faster removal rate for labelled BSA compared to IgG antibody. If FcRn-based transcytosis were an alternative means for antibodies to leave the tissue interstitial space, the FcRn non-binding antibody would have a lower washout rate than the anti-IL-17 IgG with normal FcRn binding. The measured removal rates for skin and muscle for the three macromolecules are shown in Fig. 7.

No statistically significant differences in washout of the three different tracers were found using one-way ANOVA (P values of 0.5 and 0.6 for skin and muscle, respectively), indicating no influence of molecular weight and FcRn-based transcytosis on the measured removal rate. The actual flow values were determined by multiplying the mean removal rates (Fig. 7) by the interstitial tissue volume (Table 2) as assessed earlier in the present study in accordance with eqn (5). The lymph flow derived was $0.633 \pm 0.134 \text{ ml h}^{-1}$ for skin and $0.162 \pm 0.052 \text{ ml h}^{-1}$ for muscle.

PBPK modelling

The available data and parameter values were integrated into the PBPK model and PK in the different tissue

subcompartments. This enabled estimation of the remaining biodistribution parameters for monoclonal antibodies and the model could subsequently be used for prediction of antibody PK in different tissue subcompartments following alternative dosing schedules, showing potential value as a tool for translation to other species. The final model fostered a good description of the biodistribution data and PK in the different tissue subcompartments (interstitial and vascular). Model parameters were estimated precisely. The parameter values in the final model are shown in Table 2 and 3, where Table 2 contains the tissue specific parameters and Table 3 other parameters as used in the PBPK model.

Model fits and the prediction of drug content per gram of tissue in the various tissue subcompartments for all analysed tissues are depicted in Fig. 8. For tissues with discontinuous capillaries (liver and spleen), the vascular and interstitial space cannot be discriminated because of the quick exchange in these spaces. Interstitial concentrations are expected to follow that of plasma. In tissues with very tight or size-selective capillaries (blood–brain barrier, glomerular filter of the kidneys), antibodies are expected to be restricted to the vascular space and no or negligible interstitial concentrations are predicted. In these cases, lymph flow and distribution to the tissue interstitial space cannot be determined using model estimation, and the measured interstitial volume fraction not be employed as an input value. The interstitial concentrations measured in the skin and muscle were well-described by the model featuring an estimated interstitial exclusion of 37% and 25%, respectively.

The sensitivity of simulated PK profiles on perturbations on individual parameters was evaluated. Changes in tissue-specific parameters did not significantly affect the PK in different tissues. Altering the systemic plasma clearance had a great impact on the elimination phase for all tissues. In the liver, spleen, brain and kidney, varying the vascular or lumped vascular/interstitial volume fractions strongly affected the level of the simulated PK profile. In tissues with continuous capillaries, parameter sensitivity was high for interstitial and vascular volume fractions, thereby modifying the

concentration level. Altering the lymph flow and tissue distribution flow in these tissues, on the other hand, notably affected the concentration level at the same time as having an effect on the kinetics (i.e. the time when the maximal concentration is observed). Generally, insignificant sensitivity was observed for tissue plasma flows. The calculated sensitivities for each parameter, averaged over the simulated time course, are summarized in a sensitivity matrix shown in Fig. 9.

Discussion

In the present study, we investigated determinants of tissue distribution and biophase concentrations of

therapeutic antibodies in different tissues using: (i) tissue centrifugation to directly assess interstitial concentrations and (ii) correction of total tissue concentrations and the related impact on PBPK modelling. Also, early time points (i.e. 10, 30 and 90 min) were sampled during the antibody biodistribution study, aiming to obtain insights into tissue distribution within the individual tissues.

In terms of total tissue concentrations, the results of the present study are in strong agreement with previous published antibody biodistribution studies, which can be seen first-hand when comparing the antibody biodistribution coefficients with those compiled by Shah & Betts (2013). Additionally, base distribution parameters, volume of distribution (1.25 ml) and systemic clearance

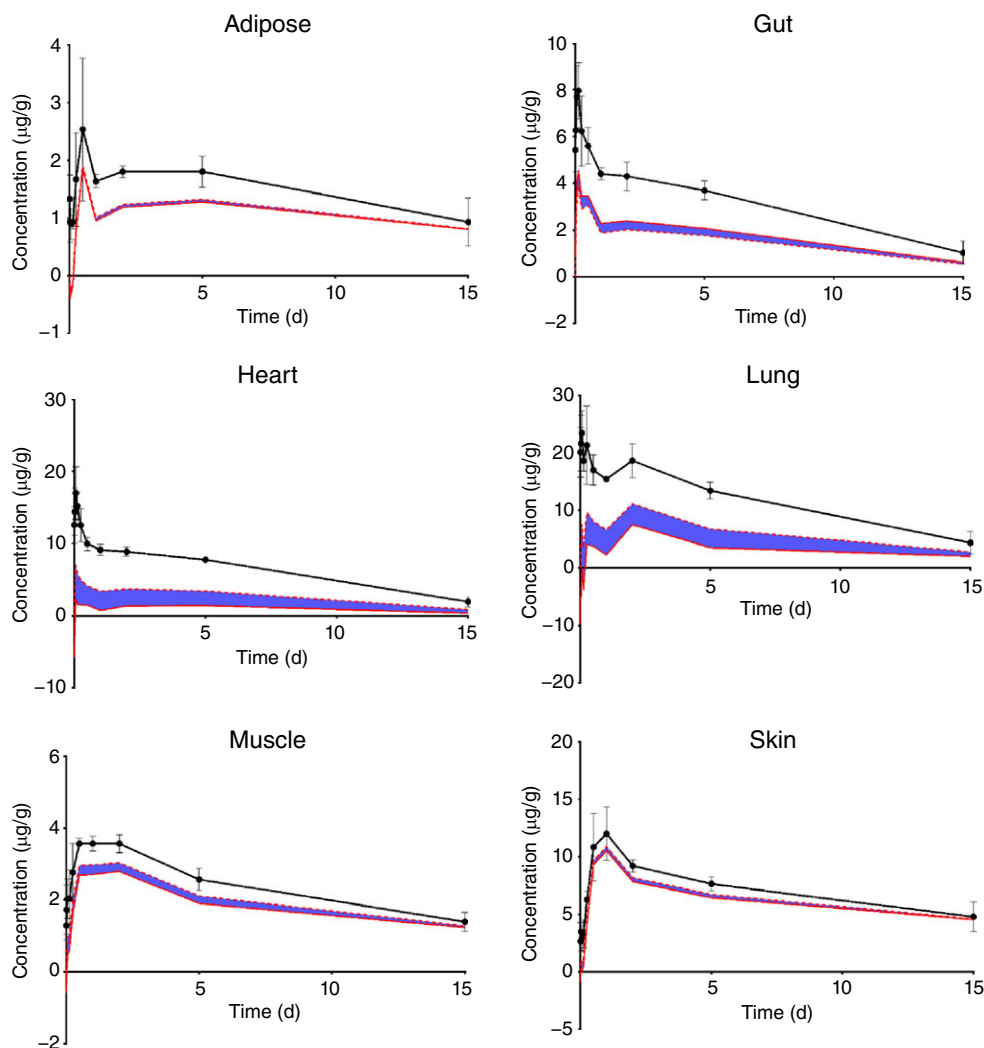


Figure 5. Total anti-IL17 antibody PK in adipose, gut, heart, lung, muscle and skin corrected for drug content in corresponding ^{125}I -albumin distribution space

Measured total tissue PK profiles (solid profile with mean $\pm$ SD) corrected for the amount of drug in residual plasma assessed as ^{125}I -albumin distribution volume (solid red curve) and anti-IL-17 IgG distribution volume (dashed red curve) after 5 and 10 min of circulation time, respectively. The blue-shaded area shows the anticipated extravascular antibody concentration after accounting for drug content in the calculated tracer distribution spaces. The size of the blue-shaded area is defined by the difference in the volume fraction assessed by ^{125}I -albumin distribution after 5 min and the ^{125}I -anti-IL-17 IgG distribution after 10 min.

($8.6 \text{ ml day}^{-1} \text{ kg}^{-1}$), were all within the expected range (Deng *et al.* 2011). Important differences are notable, however, when reviewing the expected extravascular and interstitial concentrations. In the present study, we showed a high interstitial antibody exposure to target cells in the muscle and skin with concentrations up to 50% of the plasma concentration. For large molecules, corrections

for drug in residual plasma are frequently performed to calculate extravascular concentrations. Interestingly, these corrections, with residual plasma of the respective tissues measured by ^{125}I -HSA and ^{125}I -anti-IL-17 IgG distribution space, resulted in negligible anticipated extravascular concentrations in the brain and kidney. This is probably explained by the very tight capillaries constituting the

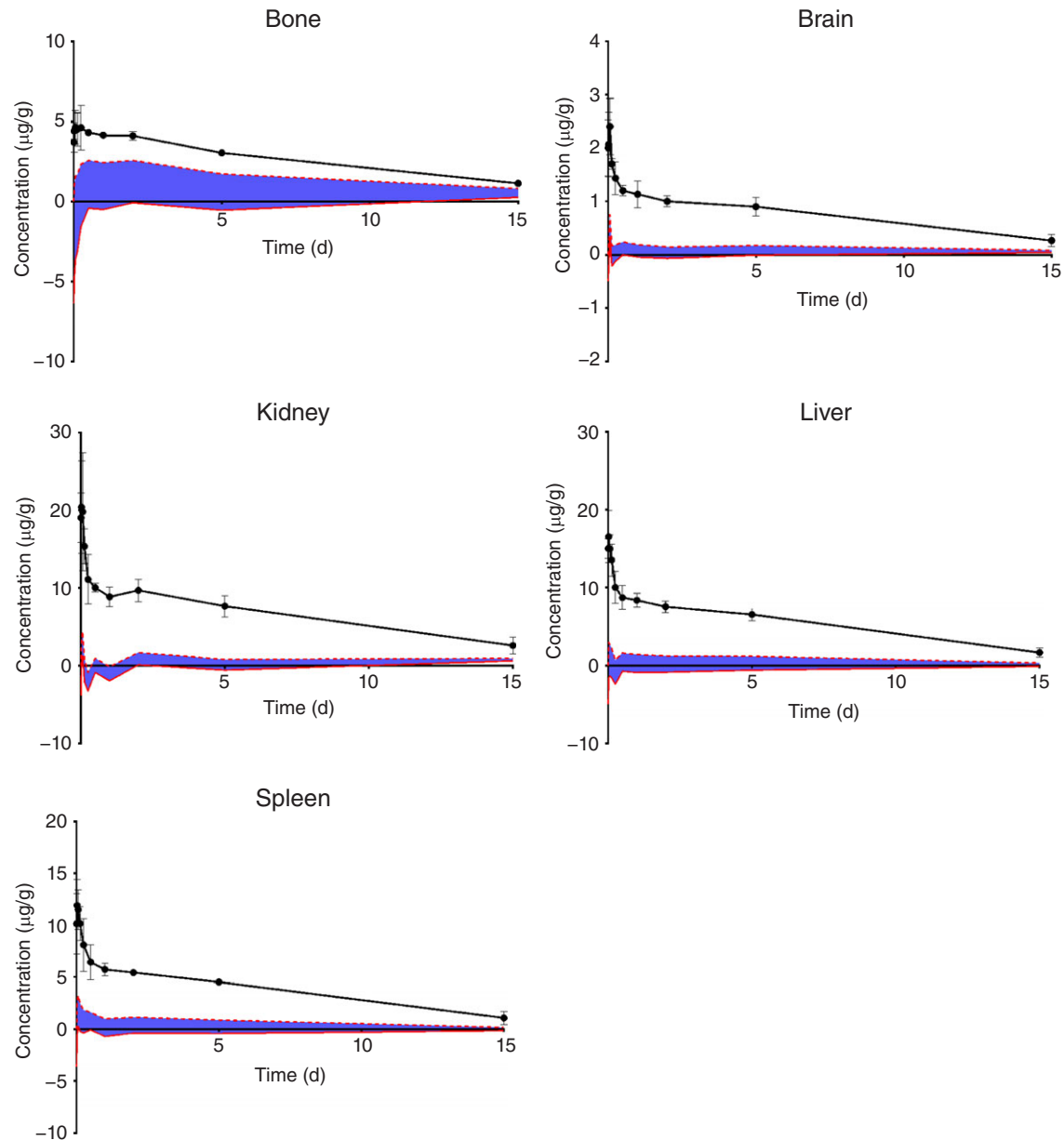


Figure 6. Total anti-IL17 antibody PK in bone, brain, kidney, liver and spleen corrected for drug content in corresponding ^{125}I -albumin distribution space

Measured total tissue PK profiles (solid profile with mean $\pm$ SD) corrected for amount of drug in residual plasma assessed as ^{125}I -albumin distribution volume (solid red curve) and anti-IL-17 IgG distribution volume (dashed red curve) after 5 and 10 min of circulation time, respectively. In these tissues, no or very low extravascular tissue concentrations were expected after accounting for the drug in the assessed tracer spaces (blue-shaded area). The size of the blue-shaded area is defined by the difference in the volume fraction assessed by ^{125}I -albumin distribution after 5 min and the ^{125}I -anti-IL-17 IgG distribution after 10 min.

blood–brain barrier and the size-selective fenestrated capillaries in the glomerular filter. Both types of capillaries practically prevent IgG antibodies from entering the tissue space. Furthermore, with regard to the liver, spleen and bone, the zero line was included after correcting for the amount of drug in the distribution space of the macromolecular tracer. For these tissues, however, the interpretation of this finding would be the opposite. These tissues possess discontinuous capillary walls that are non-restrictive to proteins (Rippe & Haraldsson, 1994; Sarin, 2010). Therefore, the measured space with the macromolecular tracer used for correction most probably represents a mix of the plasma and interstitial space. Correcting for the amount of drug in that space should therefore not be performed to calculate extravascular concentrations in such tissues. Our results actually suggest that the tracer is equilibrated in these organs at between 5 and 10 min of circulation time. As a result of this facilitated exchange with the plasma space, the extravascular space of tissues, with discontinuous capillaries, might be interpreted as extended plasma spaces. This finding leads to the assumption that interstitial concentrations in tissue with discontinuous capillaries are reflected by the plasma concentration. Overall, our results provide evidence that total tissue concentrations are much lower than the actual interstitial concentrations in tissues (except in the brain and kidneys). Therefore, our findings are in contrast to the common notion in the pharmacological literature that interstitial concentrations are generally much lower than vascular concentrations (Lobo *et al.* 2004; Wang *et al.* 2008). On the other hand, they are

in agreement with earlier published pre-nodal lymph to plasma concentration ratios of macromolecules in various tissues (Aukland & Reed, 1993; Michel & Curry, 1999), where pre-nodal lymph can be expected to be representative for interstitial fluid in steady-state conditions.

Distribution spaces for ^{125}I -HSA- and chromium EDTA were measured with robust precision. The residual plasma fractions based on HSA distribution were systematically higher than reported previously (Garg, 2007; Boswell *et al.* 2014). These previous values are based on the distribution of red blood cells. In such cases, a systemic haematocrit value is regularly employed used to derive the residual plasma volumes in tissues. Yet, it is well-established that the local haematocrit in the smaller tissue vessels might be considerably lower than the systemic haematocrit, known as the 'Fåhræus Lindqvist effect' or the 'screening effect' (Goldsmith *et al.* 1989; Fung, 1993). Therefore, utilizing systemic haematocrit values to derive residual plasma fractions based on red blood cell fractions will erroneously lead to lower correction factors, thereby introducing a systematic bias. The quantitative importance of such bias is shown in Fig. 10, where there is the comparison of the PK profiles for the analysed tissues, including residual plasma corrections based on red blood cell distribution (Garg, 2007; Boswell *et al.* 2014) and based on ^{125}I -HSA and ^{125}I -anti-IL-17 IgG distribution after 5 and 10 min of circulation time, respectively. Hence, a direct assessment of the plasma space is, in our opinion, more representative, with the exception of organs with discontinuous capillaries, where the tracer will quickly extravasate and equilibrate in the entire extravascular phase as discussed earlier. Extracellular volume fractions measured by ^{51}Cr -EDTA distribution are also in solid agreement with the results of previous studies (Pierson *et al.* 1978; Tsuji *et al.* 1983; Boswell *et al.* 2014), especially considering the different properties of the applied tracers. These findings show that corrective factors have a major impact on estimated extravascular concentration, thus demonstrating the significance of measurements that are exact as possible, as well as careful interpretation of the results. Accordingly, a critical evaluation of the used tracer is warranted with such values as fixed input parameters in a PBPK model.

Lymph flow plays an important role in the biodistribution of therapeutic antibodies because of its involvement in the transport of filtered macromolecules from the interstitium back to the systemic blood circulation (Wiig & Swartz, 2012). As such, it is a critical parameter in most large molecule PBPK models. However, the input values used for tissue lymph flow in different published PBPK models vary by a factor of up to ~5000-fold and no systematic measurement values are available (Ferl *et al.* 2005; Fronton *et al.* 2014). In the present study, we derived tissue lymph flow

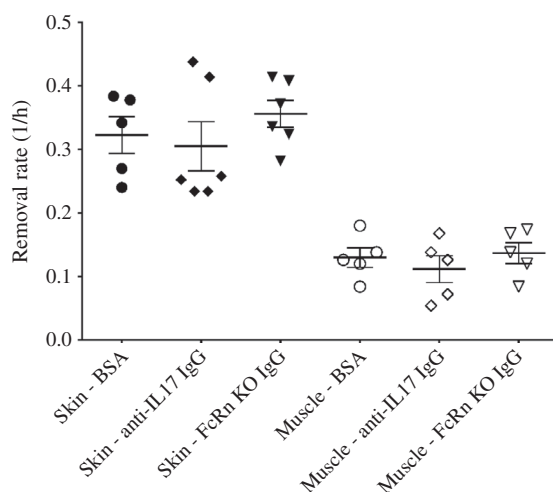


Figure 7. Lymphatic removal rate in skin and muscle assessed by fluorescent tracer removal

Measured removal rates in skin (filled) and muscle (open) for BSA, normal IgG antibody and FcRn non-binding IgG, respectively. Washout rates were 0.32 ± 0.06 , 0.31 ± 0.09 and 0.36 ± 0.05 l h^{-1} in skin and 0.13 ± 0.03 , 0.11 ± 0.05 and 0.14 ± 0.04 l h^{-1} in muscle for BSA, IgG and FcRn non-binding IgG, respectively.

Table 2. Summary of the tissue-specific parameters used and estimated for the PBPK modelling approach

	f_{ResPla} (ml g ⁻¹) ^a	Tissue distribution (ml h ⁻¹) ^b	Lymph flow (ml h ⁻¹) ^b	Frct. CL ^c	Organ volume (ml) ^d	f_{Vint} (ml g ⁻¹) ^e	Plasma flow (ml h ⁻¹) ^d
Adipose	0.006 ($\pm 1.7 \times 10^{-5}$)	0.001 ($\pm 5.9\text{e-}6$)	0.005 ($\pm 3.9 \times 10^{-5}$)	0.027	1.775	0.093	11.7
Bone	0.024 ($\pm 7.9 \times 10^{-5}$)	0.007 ($\pm 9.9 \times 10^{-5}$)	0.041 ($\pm 5.6 \times 10^{-4}$)	0.100	2.525	0.133	13.3
Brain	0.012 ($\pm 1.7 \times 10^{-5}$)	–	–	0.001	0.425	–	10.4
Gut	0.023 ($\pm 1.1 \times 10^{-4}$)	0.031 ($\pm 2.0 \times 10^{-4}$)	0.212 ($\pm 7.2 \times 10^{-4}$)	0.026	1.050	0.175	65.4
Heart	0.070 ($\pm 1.7 \times 10^{-4}$)	0.004 ($\pm 9.1 \times 10^{-5}$)	0.020 ($\pm 4.1 \times 10^{-4}$)	0.005	0.135	0.123	31.8
Kidney	0.101 ($\pm 1.3 \times 10^{-4}$)	–	–	0.060	0.470	–	59.6
Liver	0.082 ($\pm 1.1 \times 10^{-4}$)	–	–	0.300	1.725	–	9.1
Lung	0.118 ($\pm 2.7 \times 10^{-5}$)	0.002 ($\pm 2.4 \times 10^{-5}$)	0.005 ($\pm 5.7 \times 10^{-5}$)	0.013	0.183	0.227	324
Muscle	0.007 ($\pm 2.7 \times 10^{-5}$)	0.034 ($\pm 1.5 \times 10^{-4}$)	0.124 ^f ($\pm 5.7 \times 10^{-4}$)	0.190	10.1	0.130	74.8
Skin	0.014 ($\pm 8.3 \times 10^{-5}$)	0.107 ($\pm 1.6 \times 10^{-4}$)	0.467 ^f ($\pm 3.7 \times 10^{-4}$)	0.250	4.475	0.430	24.3
Spleen	0.061 ($\pm 8.7 \times 10^{-5}$)	–	–	0.028	0.125	–	7.1

^aEstimated but restricted to the range of the measured values $\pm 20\%$. ^bEstimated values. ^cFixed values based on (Eigenmann *et al.* 2017), note: bone and adipose assumed. ^dRelative values derived from Shah & Betts (2012) and scaled to a 25-g mouse. ^eFixed to measured values. ^fEstimated but restricted to the range of the measured values $\pm 30\%$. The 95% confidence intervals (CI) for the parameters estimates are defined in brackets below the estimated value with the upper and lower bound of the 95% CI being defined as the estimate + or – the value in the bracket, respectively. A narrow confidence interval indicates a precise estimation of the respective model parameters. f_{ResPla} , residual plasma fraction; Frct. CL, fractional contribution to total systemic clearance; f_{Vint} , interstitial volume fraction.

Table 3. Summary of other parameters used and estimated for the PBPK modelling approach

Syst. V_{pla} (ml) ^a	Total CL (ml h ⁻¹) ^a	CO plasma (ml h ⁻¹) ^b	Frct. access. skin ^a	Frct. access. muscle ^a	f_{ResPla} skin cent. ^c	f_{ResPla} muscle cent. ^c
0.793 ($\pm 7.9 \times 10^{-5}$)	0.009 ($\pm 2.9 \times 10^{-5}$)	324	0.634 (± 0.001)	0.748 (± 0.002)	0.065	0.056

^aEstimated values. ^bRelative values derived from Shah & Betts (2012) and scaled to a 25-g mouse. ^cEstimated but restricted to the range of the measured values $\pm 20\%$. The 95% confidence intervals (CI) for the parameters estimates are defined in brackets below the estimated value with the upper and lower bound of the 95% CI being defined as the estimate + or the value in the bracket, respectively. A narrow confidence interval indicates a precise estimation of the respective model parameters. Syst. V_{pla} , systemic plasma volume; Total CL, total systemic clearance; CO plasma, plasma cardiac output; Frct. Access, accessible volume fraction in interstitial space; f_{ResPla} cent., residual plasma fraction in tissue centrifugate.

values in the muscle and skin based on macromolecular washout involving directly use measured values in PBPK models. Integration of both measured lymph flow values results in an adequate description of the biodistribution data. Optimized lymph flow values by model-based parameter estimation would, however, be lower in the skin (0.15 vs. 0.47 ml h⁻¹) and slightly lower in muscle

(0.10 vs. 0.12 ml h⁻¹). Nevertheless, measured lymph flows serve as reliable input values for the PBPK model and foster a critical evaluation of previously used values. This also suggests that flow is highly tissue-specific and might not be well-captured by fixing it to a given fraction of the respective plasma flow for all tissues. The lymph flow values that we employed based on our *in vivo* washout

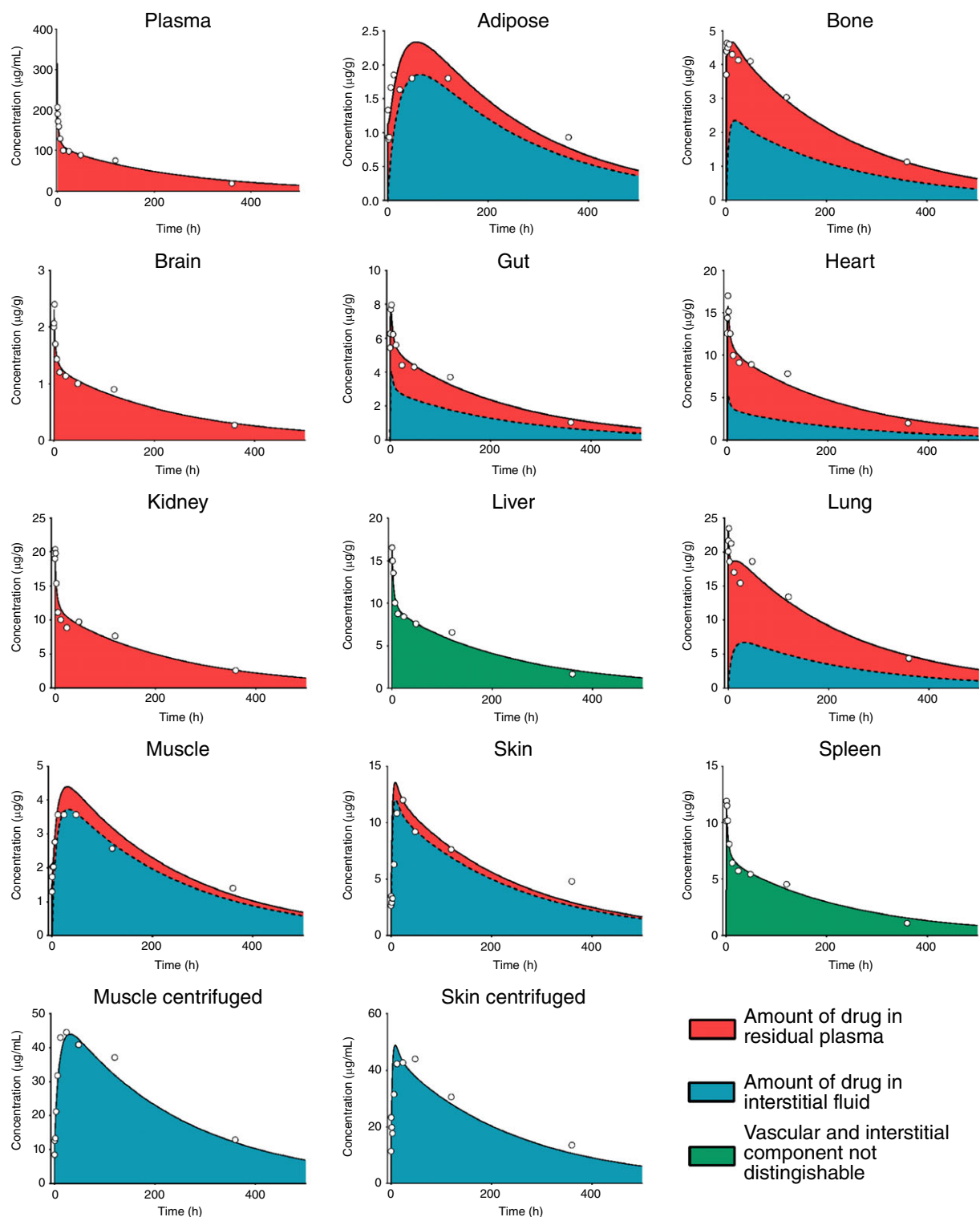
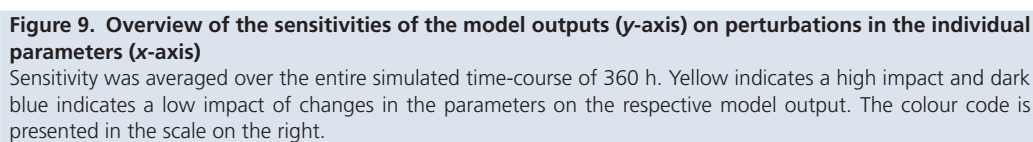


Figure 8. PBPK model prediction of antibody PK in residual plasma and interstitial space of individual tissues

Model description of the biodistribution data. The model prediction is shown as the black continuous line. The fitted mean PK data are represented by the open circles. The amount of drug in the residual plasma of the tissues is depicted in red, whereas the blue area delineated by the black dashed line indicates the amount of drug in the interstitial fluid. The area is colored in green when interstitial space and plasma cannot be distinguished.

The PBPK modelling approach showed that, for tissues with very limited or very quick extravasation, estimation of parameters for antibody distribution to or removal from the interstitial space is not feasible based on the biodistribution data, despite extensive and early sampling times. The model structure therefore did not include distribution to and removal from the interstitial space for these tissues and either antibody distribution was restricted to the vascular space in the case of very tight capillary structures or the plasma and interstitial space were lumped as one extracellular distribution space in tissues with discontinuous capillaries. For the tissues with discontinuous capillaries, the biophase concentration would hence follow the plasma concentration levels. The amount of drug measured in the brain and kidney was explainable solely by the expected drug in the residual plasma of these tissues. It should be noted that, if corrective factors for residual plasma are fixed values that are too low, the model would be forced to describe the remaining unexplained antibody content by estimation of distribution to and removal from the interstitial space. For tissues with continuous capillaries, measured interstitial and residual plasma fractions were used and a tissue distribution flow was estimated in the model, allowing a detailed description of the biodistribution data. Overall, this is a relatively simple PBPK modelling approach compared to many of the more detailed previously published PBPK models (Ferland *et al.* 2005; Garg & Balthasar, 2007; Urva *et al.* 2010; Chen & Balthasar, 2012). It is worth noting, however, our intention to keep the model complexity rather

In conclusion, we have provided novel data essential for PBPK modelling of monoclonal antibodies. To our knowledge, this is the first time that a PBPK model has been used to directly describe measured tissue interstitial PK for monoclonal antibodies. The measured input parameters, their direct integration and the critical evaluation of the measured and previously used values, as well as the underlying experimental methods, are vital for a more realistic interpretation of tissue PK data for antibodies. This allows us to model antibody tissue distribution and removal in a more tissue-specific way based on physiological rationale. We show that the use of residual plasma correction factors based on red blood cell distribution can lead to errors in estimation of extravascular antibody concentrations. Most importantly, we demonstrate by direct measurements and model-based data interpretation that, in most tissues, high interstitial concentrations can be achieved for antibodies, which contradicts the still-prevailing view that antibody



distribution to the tissue interstitial space is quite limited and low concentrations are expected. These findings also provide new insights with a potential impact on the development of IgG therapeutics. They show that the use of total tissue concentrations of antibodies can be highly misleading and does not reflect how much of the therapeutic antibody potentially reaches its target in the interstitial space. Correlating this concentration to a therapeutic effect could therefore bias the estimated

potency of the respective antibody drug, and any potential *in vitro*–*in vivo* correlation or upfront predictions of drug effect would be biased. The model developed in the present study offers a more precise prediction of the distribution of IgG antibodies to the interstitial space of individual tissues. These predicted target-site concentrations might be utilized to better assess the PKPD relationship and potency of antibody therapeutics. This knowledge could then potentially be further employed to more accurately

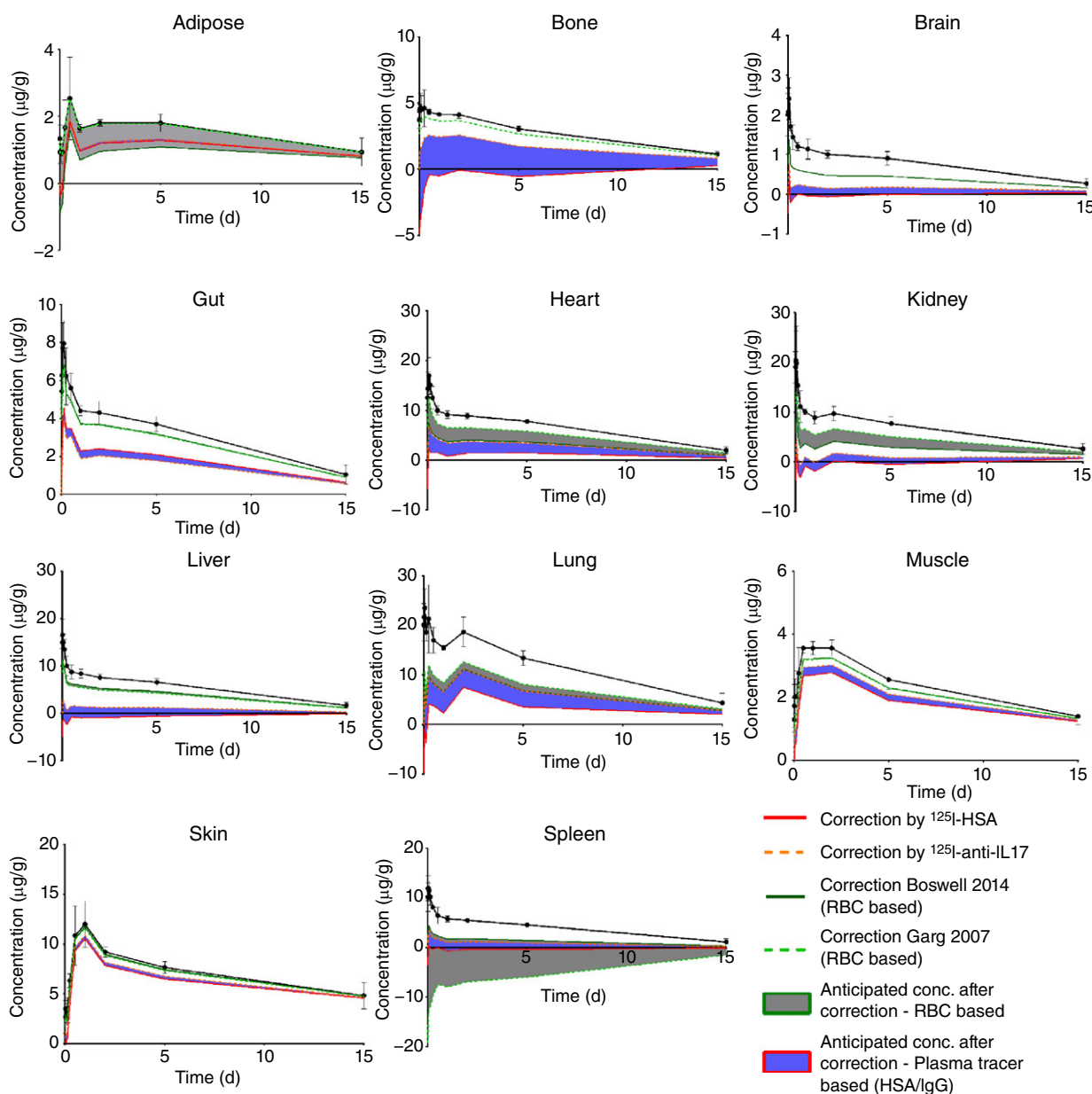


Figure 10. Total tissue PK in tissues corrected for amount of drug in residual plasma in harvested tissues

Residual plasma volumes are determined based on either red blood cell distribution volumes reported by Garg (2007) green-dashed) and Boswell *et al.* (2014) (green-solid) and converted to plasma using a haematocrit value of 0.45 or by directly assessing the plasma space with labelled HSA (solid red) and IgG (dashed orange). It is clearly demonstrated that the red blood cell-based method leads systematically to a lower correction and thus higher expected extravascular concentrations in tissues (grey-shaded area) compared to when directly assessed using plasma tracers (blue-shaded area).

predict expected PK and the therapeutic effect of IgG antibody agents upfront, also following alternative dosing schedules.

We consider that our findings are broadly applicable for IgG monoclonal antibodies that currently are the primary isotype for therapeutic antibodies. However, we recognize that differences in physicochemical properties (e.g. hydrodynamic radius or charge) for other Ig isotypes or engineered antibodies might affect certain distribution properties of protein therapeutics (e.g. passage through the endothelial layer or exclusion volume in the interstitial space). Investigating the impact of such factors on specific PK parameters during future studies, specifically combining experiments with PBPK modelling, could ultimately comprise a valuable next step with respect to improving PBPK modelling for antibody therapeutics.

Appendix: Parameter definitions

A_{tis} = amount in tissue	A_{tis} = amount in tissue
$A_{v_{tis}}$ = amount in tissue residual plasma	i_{tis} = amount in tissue interstitial space
$C_{int_{tis}}$ = tissue interstitial concentration	C_{tis} = total tissue concentration
fCL_{tis} = fractional contribution to clearance	CL_{intpla} = total systemic clearance
Q_{tis} = tissue arterial plasma flow	L_{tis} = tissue lymph flow
V_{pla} = systemic plasma volume	$Inflow$ = antibody tissue distribution flow
V_{tis} = tissue interstitial volume	V_{tis} = total tissue volume
	$V_{v_{tis}}$ = tissue residual plasma volume

Model differential equations

Plasma.

$$\begin{aligned} \frac{dA_{pla}}{dt} = & \frac{L_{bon} \times A_{i_{bon}}}{V_{i_{bon}}} - \frac{Q_{bon} \times A_{pla}}{V_{pla}} + \frac{(Q_{lun} - L_{lun}) \times A_{v_{lun}}}{V_{v_{lun}}} \\ & + \frac{L_{lun} \times A_{i_{lun}}}{V_{i_{lun}}} - \frac{Q_{a_{liv}} \times A_{pla}}{V_{pla}} - \frac{Q_{ski} \times A_{pla}}{V_{pla}} \\ & + \frac{L_{ski} \times A_{i_{ski}}}{V_{i_{ski}}} - \frac{Q_{mus} \times A_{pla}}{V_{pla}} + \frac{L_{mus} \times A_{i_{mus}}}{V_{i_{mus}}} \\ & - \frac{Q_{spl} \times A_{pla}}{V_{pla}} - \frac{Q_{gut} \times A_{pla}}{V_{pla}} + \frac{L_{gut} \times A_{i_{gut}}}{V_{i_{gut}}} \\ & - \frac{Q_{kid} \times A_{pla}}{V_{pla}} - \frac{Q_{hea} \times A_{pla}}{V_{pla}} + \frac{L_{hea} \times A_{i_{hea}}}{V_{i_{hea}}} \\ & - \frac{Q_{adi} \times A_{pla}}{V_{pla}} - \frac{Q_{bra} \times A_{pla}}{V_{pla}} + \frac{L_{adi} \times A_{i_{adi}}}{V_{i_{adi}}} \end{aligned}$$

Adipose. Vascular:

$$\begin{aligned} \frac{dA_{v_{adi}}}{dt} = & - \frac{Inflow_{adi} \times A_{v_{adi}}}{V_{v_{adi}}} + \frac{Q_{adi} \times A_{pla}}{V_{pla}} \\ & - \frac{(Q_{adi} - L_{adi}) \times A_{v_{adi}}}{V_{v_{adi}}} - \frac{A_{v_{adi}}}{V_{v_{adi}}} \\ & \times CL_{intpla} \times fCL_{adi} \end{aligned}$$

Interstitial:

$$\frac{dA_{i_{adi}}}{dt} = \frac{Inflow_{adi} \times A_{v_{adi}}}{V_{v_{adi}}} - \frac{L_{adi} \times A_{i_{adi}}}{V_{i_{adi}}}$$

Bone. Vascular:

$$\frac{dA_{i_{bon}}}{dt} = \frac{Inflow_{bon} \times A_{v_{bon}}}{V_{v_{bon}}} - \frac{L_{bon} \times A_{i_{bon}}}{V_{i_{bon}}}$$

Interstitial:

$$\begin{aligned} \frac{dA_{v_{bon}}}{dt} = & - \frac{Inflow_{bon} \times A_{v_{bon}}}{V_{v_{bon}}} + \frac{Q_{bon} \times A_{pla}}{V_{pla}} \\ & - \frac{(Q_{bon} - L_{bon}) \times A_{v_{bon}}}{V_{v_{bon}}} - \frac{A_{v_{bon}}}{V_{v_{bon}}} \\ & \times CL_{intpla} \times fCL_{bon} \end{aligned}$$

Brain.

$$\begin{aligned} \frac{dA_{bra}}{dt} = & \frac{Q_{bra} \times A_{pla}}{V_{pla}} - \frac{Q_{bra} \times A_{v_{bra}}}{V_{v_{bra}}} - \frac{A_{v_{bra}}}{V_{v_{bra}}} \\ & \times CL_{intpla} \times fCL_{bra} \end{aligned}$$

Gut. Vascular:

$$\begin{aligned} \frac{dA_{p_{gut}}}{dt} = & - \frac{Inflow_{gut} \times A_{v_{gut}}}{V_{v_{gut}}} + \frac{Q_{gut} \times A_{pla}}{V_{pla}} \\ & - \frac{(Q_{gut} - L_{gut}) \times A_{v_{gut}}}{V_{v_{gut}}} - \frac{A_{v_{gut}}}{V_{v_{gut}}} \\ & \times CL_{intpla} \times fCL_{gut} \end{aligned}$$

Interstitial:

$$\frac{dA_{i_{gut}}}{dt} = \frac{Inflow_{gut} \times A_{v_{gut}}}{V_{v_{gut}}} - \frac{L_{gut} \times A_{i_{gut}}}{V_{i_{gut}}}$$

Heart. Vascular:

$$\begin{aligned} \frac{dA_{v_{hea}}}{dt} = & \frac{Q_{hea} \times A_{pla}}{V_{pla}} - \frac{Inflow_{hea} \times A_{v_{hea}}}{V_{v_{hea}}} \\ & - \frac{(Q_{hea} - L_{hea}) \times A_{v_{hea}}}{V_{v_{hea}}} - \frac{A_{v_{hea}}}{V_{v_{hea}}} \\ & \times CL_{intpla} \times fCL_{hea} \end{aligned}$$

Interstitial:

$$\frac{dA_{i_{hea}}}{dt} = \frac{Inflow_{hea} \times A_{v_{hea}}}{V_{v_{hea}}} - \frac{L_{hea} \times A_{i_{hea}}}{V_{i_{hea}}}$$

Kidney.

$$\begin{aligned} \frac{dA_{kid}}{dt} = & \frac{Q_{kid} \times A_{pla}}{V_{pla}} - \frac{Q_{kid} \times A_{v_{kid}}}{V_{v_{kid}}} - \frac{A_{v_{kid}}}{V_{v_{kid}}} \\ & \times CL_{intpla} \times fCL_{kid} \end{aligned}$$

Liver.

$$\begin{aligned} \frac{dA_{\text{liv}}}{dt} = & \frac{Q_{\text{a}} \times A_{\text{pla}}}{V_{\text{pla}}} - \frac{(Q_{\text{a}} + Q_{\text{gut}} - L_{\text{gut}} + Q_{\text{spl}}) \times A_{\text{liv}}}{V_{\text{d}_{\text{liv}}}} \\ & + \frac{Q_{\text{spl}} \times A_{\text{spl}}}{V_{\text{d}_{\text{spl}}}} + \frac{(Q_{\text{gut}} - L_{\text{gut}}) \times A_{\text{v}_{\text{gut}}}}{V_{\text{v}_{\text{gut}}}} \\ & - \frac{A_{\text{v}_{\text{liv}}}}{V_{\text{d}_{\text{liv}}}} \times CL_{\text{intpla}} \times fCL_{\text{liv}} \end{aligned}$$

Lung. Vascular:

$$\begin{aligned} \frac{dA_{\text{v}_{\text{lun}}}}{dt} = & - \frac{Inflow_{\text{lun}} \times A_{\text{v}_{\text{lun}}}}{V_{\text{v}_{\text{lun}}}} - \frac{(Q_{\text{lun}} - L_{\text{lun}}) \times A_{\text{v}_{\text{lun}}}}{V_{\text{v}_{\text{lun}}}} \\ & + \frac{(Q_{\text{bon}} - L_{\text{bon}}) \times A_{\text{v}_{\text{bon}}}}{V_{\text{v}_{\text{bon}}}} \\ & + \frac{(Q_{\text{liv}} + Q_{\text{gut}} - L_{\text{gut}} + Q_{\text{spl}}) \times A_{\text{liv}}}{V_{\text{v}_{\text{liv}}}} \\ & + \frac{(Q_{\text{ski}} - L_{\text{ski}}) \times A_{\text{v}_{\text{ski}}}}{V_{\text{v}_{\text{ski}}}} + \frac{(Q_{\text{mus}} - L_{\text{mus}}) \times A_{\text{v}_{\text{mus}}}}{V_{\text{v}_{\text{mus}}}} \\ & + \frac{Q_{\text{kid}} \times A_{\text{v}_{\text{kid}}}}{V_{\text{v}_{\text{kid}}}} + \frac{(Q_{\text{hea}} - L_{\text{hea}}) \times A_{\text{v}_{\text{hea}}}}{V_{\text{v}_{\text{hea}}}} \\ & + \frac{(Q_{\text{adi}} - L_{\text{adi}}) \times A_{\text{v}_{\text{adi}}}}{V_{\text{v}_{\text{adi}}}} + \frac{Q_{\text{bra}} \times A_{\text{v}_{\text{bra}}}}{V_{\text{v}_{\text{bra}}}} \\ & - \frac{A_{\text{v}_{\text{lun}}}}{V_{\text{v}_{\text{lun}}}} \times CL_{\text{intpla}} \times fCL_{\text{lun}} \end{aligned}$$

Interstitial:

$$\frac{dA_{i_{\text{lun}}}}{dt} = \frac{Inflow_{\text{lun}} \times A_{\text{v}_{\text{lun}}}}{V_{\text{v}_{\text{lun}}}} - \frac{L_{\text{lun}} \times A_{i_{\text{lun}}}}{V_{i_{\text{lun}}}}$$

Muscle. Vascular:

$$\begin{aligned} \frac{dA_{\text{v}_{\text{mus}}}}{dt} = & \frac{Q_{\text{mus}} \times A_{\text{pla}}}{V_{\text{pla}}} - \frac{Inflow_{\text{mus}} \times A_{\text{v}_{\text{mus}}}}{V_{\text{v}_{\text{mus}}}} \\ & - \frac{(Q_{\text{mus}} - L_{\text{mus}}) \times A_{\text{v}_{\text{mus}}}}{V_{\text{v}_{\text{mus}}}} - \frac{A_{\text{v}_{\text{mus}}}}{V_{\text{v}_{\text{mus}}}} \\ & \times CL_{\text{intpla}} \times fCL_{\text{mus}} \end{aligned}$$

Interstitial:

$$\frac{dA_{i_{\text{mus}}}}{dt} = \frac{Inflow_{\text{mus}} \times A_{\text{v}_{\text{mus}}}}{V_{\text{v}_{\text{mus}}}} - \frac{L_{\text{mus}} \times A_{i_{\text{mus}}}}{V_{i_{\text{mus}}}}$$

Skin. Vascular:

$$\frac{dA_{i_{\text{ski}}}}{dt} = \frac{Inflow_{\text{ski}} \times A_{\text{v}_{\text{ski}}}}{V_{\text{v}_{\text{ski}}}} - \frac{L_{\text{ski}} \times A_{i_{\text{ski}}}}{V_{i_{\text{ski}}}}$$

Interstitial:

$$\begin{aligned} \frac{dA_{\text{v}_{\text{ski}}}}{dt} = & \frac{Q_{\text{ski}} \times A_{\text{pla}}}{V_{\text{pla}}} - \frac{Inflow_{\text{ski}} \times A_{\text{v}_{\text{ski}}}}{V_{\text{v}_{\text{ski}}}} \\ & - \frac{(Q_{\text{ski}} - L_{\text{ski}}) \times A_{\text{v}_{\text{ski}}}}{V_{\text{v}_{\text{ski}}}} - \frac{A_{\text{v}_{\text{ski}}}}{V_{\text{v}_{\text{ski}}}} \\ & \times CL_{\text{intpla}} \times fCL_{\text{ski}} \end{aligned}$$

Spleen.

$$\begin{aligned} \frac{dA_{\text{spl}}}{dt} = & \frac{Q_{\text{spl}} \times A_{\text{pla}}}{V_{\text{pla}}} - \frac{Q_{\text{spl}} \times A_{\text{spl}}}{V_{\text{d}_{\text{spl}}}} - \frac{A_{\text{spl}}}{V_{\text{d}_{\text{spl}}}} \\ & \times CL_{\text{intpla}} \times fCL_{\text{spl}} \end{aligned}$$

Model definitions

$$C_{\text{pla}} = \frac{A_{\text{pla}}}{V_{\text{pla}}}$$

$$C_{\text{tis}} = \frac{A_{i_{\text{tis}}} + A_{\text{v}_{\text{tis}}}}{V_{\text{tis}}}$$

$$C_{\text{int}_{\text{ski}}} = \frac{\frac{A_{i_{\text{ski}}} + A_{\text{p}_{\text{ski}}}}{V_{\text{v}_{\text{ski}}}} \times fV_{\text{pla}_{\text{cent}_{\text{ski}}}}}{V_{i_{\text{ski}}} \times f_{\text{access}_{\text{ski}}}}$$

$$C_{\text{int}_{\text{mus}}} = \frac{\frac{A_{i_{\text{mus}}} + A_{\text{p}_{\text{mus}}}}{V_{\text{v}_{\text{mus}}}} \times fV_{\text{pla}_{\text{cent}_{\text{mus}}}}}{V_{i_{\text{mus}}} \times f_{\text{access}_{\text{mus}}}}$$

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Additional information

Competing interests

Ludivine Fronton is now employed at Bayer AG. The findings and conclusions of the present study are those of the authors and do not necessarily represent the view of Bayer AG. Other than this, the authors declare that they have no competing interests in relation to the reported work.

Author contribution

MJE, LF, MBO and HW contributed to the design and conception of the work. MJE, TVK, OT and HW contributed to the acquisition of the data. MJE, TVK, BK, OT and HW contributed to the analysis of the data. MJE, TVK, BK, OT, MBO and HW contributed to the interpretation of the data. MJE, TVK, BK, OT, LF, MBO and HW revised the work for important intellectual content. MJE, TVK and HW drafted the work. All authors approved the final version of the manuscript and agreed to be accountable for all aspects of the work to ensure that questions related to the accuracy or integrity of any part of it are appropriately investigated and resolved. All authors designated as authors qualify for authorship and those who are eligible for authorship are listed.

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Supporting information

The following supporting information is available in the online version of this article.

Data S1. Matlab script which needs to be executed in order to generate the PBPK model simulations based on the herein presented article. Note: Data S2–4 are required to be saved in the same folder in order to perform the simulations.

Data S2. Matlab script needed in order to extract the parameter values as presented in this work. Note: This file needs to be saved in the same folder as Data S1 but only Data S1 needs to be executed in Matlab.

Data S3. Matlab script required to generate the model predictions according to the given doses. Note: This file needs to be saved in the same folder as Data S1 but only Data S1 needs to be executed in Matlab.

Data S4. Simbiology PBPK model source file. Note: This file needs to be saved in the same folder as Data S1 but only Data S1 needs to be executed in Matlab. A Simbiology toolbox license is required to run the predictions in Matlab.